

Digital Logic Circuits (Spring 2026)

HW #3 (Due: April 22, 2026)

14. Implement a logic circuit for the truth table in Table 5–9.

TABLE 5–9				
Inputs				Output
A	B	C	D	X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

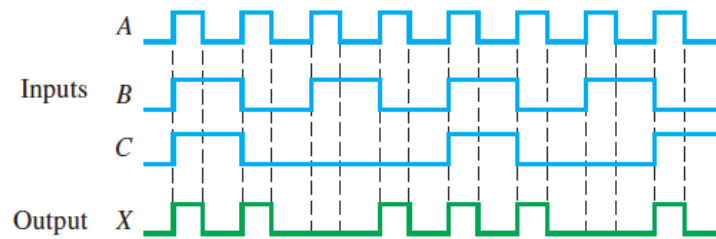
18. Minimize the gates required to implement the functions in each part of Problem 12 in SOP form.

(a) $X = \overline{A}B + CD + (\overline{A + B})(ACD + \overline{B}E)$

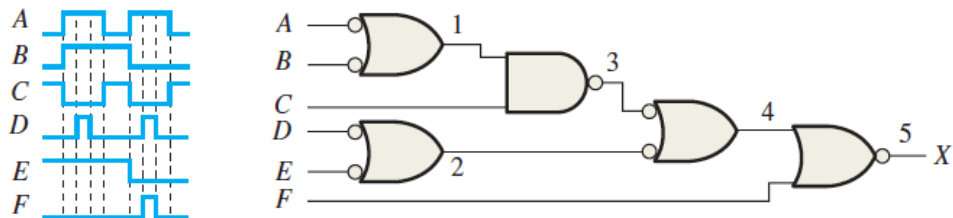
(b) $X = ABC\overline{D} + \overline{D}EF + \overline{A}F$

(c) $X = \overline{A}[B + \overline{C}(D + E)]$

30. For the input waveforms in Figure 5-61, what logic circuit will generate the output waveform shown?



32. For the circuit in Figure 5-63, draw the waveforms at the numbered points in the proper relationship to each other.



8. The input waveforms in Figure 6-72 are applied to a 2-bit adder. Determine the waveforms for the sum and the output carry in relation to the inputs by constructing a timing diagram.

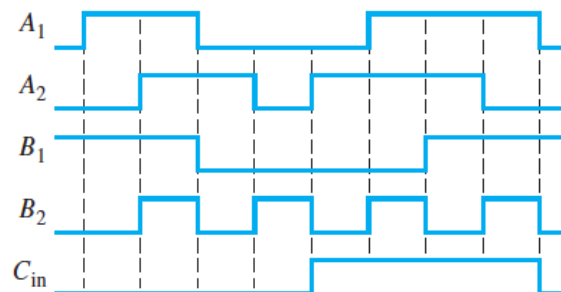
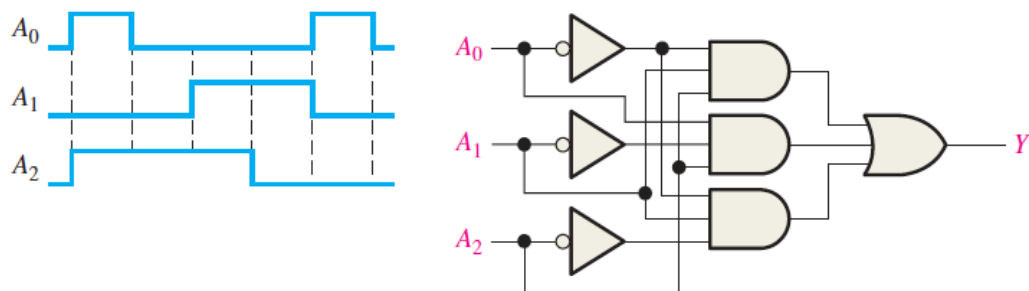


FIGURE 6-72

9. The following sequences of bits (right-most bit first) appear on the inputs to a 4-bit parallel adder. Determine the resulting sequence of bits on each sum output.

A ₁	1010
A ₂	1100
A ₃	0101
A ₄	1101
B ₁	1001
B ₂	1011
B ₃	0000
B ₄	0001

19. You wish to detect only the presence of the codes 1010, 1100, 0001, and 1011. An active-HIGH output is required to indicate their presence. Develop the minimum decoding logic with a single output that will indicate when any one of these codes is on the inputs. For any other code, the output must be LOW.
20. If the input waveforms are applied to the decoding logic as indicated in Figure 6–76, sketch the output waveform in proper relation to the inputs.



24. For the circuit in Figure 7–89, determine the maximum frequency of the clock signal for reliable operation if the set-up time for each flip-flop is 3 ns and the propagation delays (t_{PLH} and t_{PHL}) from clock to output are 6 ns for each flip-flop.

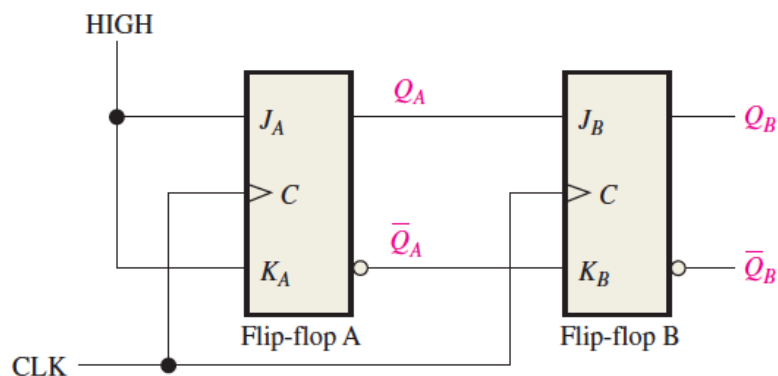


FIGURE 7–89

14. Implement a logic circuit for the truth table in Table 5-9.

TABLE 5-9				
Inputs				Output
A	B	C	D	X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0
1	1	1	1	1

18. Minimize the gates required to implement the functions in each part of Problem 12 in SOP form.

(a) $X = \bar{A}B + CD + (\bar{A} + \bar{B})(ACD + \bar{B}E)$

(b) $X = ABC\bar{D} + \bar{D}EF + \bar{A}F$

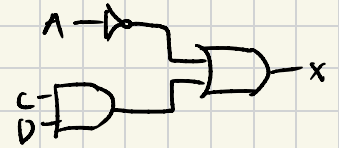
(c) $X = \bar{A}[B + \bar{C}(D + E)]$

(a) $X = \bar{A}B + CD + \bar{A}\bar{B}(ACD + \bar{B}E)$

$= \bar{A}B + CD + 0 + \bar{A}\bar{B} + \bar{A}\bar{B}E$

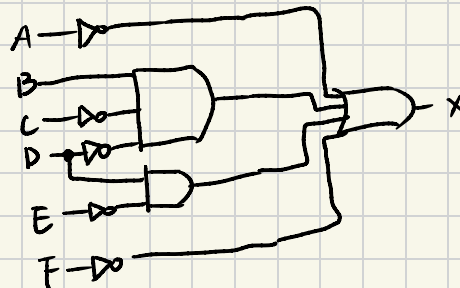
$= \bar{A} + CD + \bar{A}\bar{B}E$

$= \bar{A} + CD$

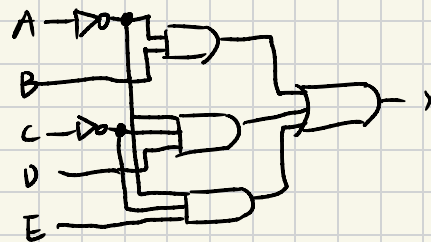


(b) $X = ABC\bar{D} + \bar{D}EF + \bar{A} + \bar{F}$

$= \bar{A} + \bar{B}\bar{C}\bar{D} + \bar{F} + \bar{D}E$



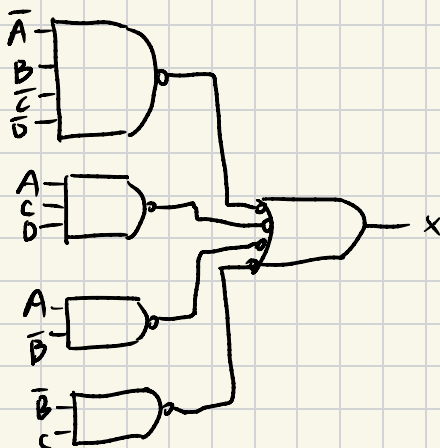
(c) $X = \bar{A}B + \bar{A}\bar{C}D + \bar{A}\bar{C}E$



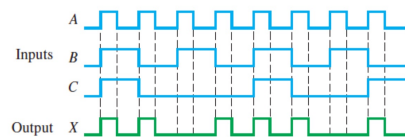
14.

CD \ AB	00	01	11	10
00	0	0	1	1
01	1	0	0	0
11	0	0	1	0
10	1	1	1	1

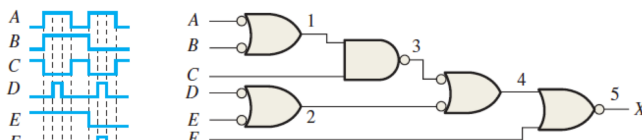
$X = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B} + ACD + \bar{B}C$



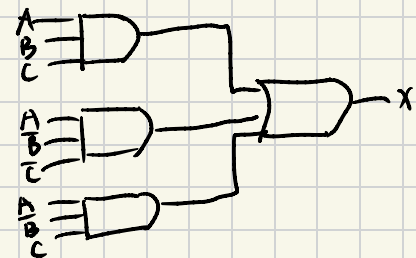
30. For the input waveforms in Figure 5-61, what logic circuit will generate the output waveform shown?



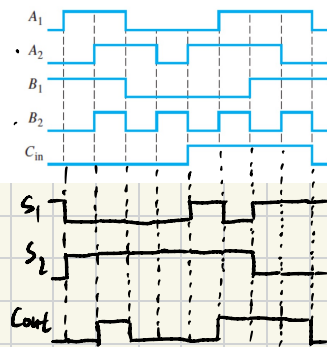
32. For the circuit in Figure 5-63, draw the waveforms at the numbered points in the proper relationship to each other.



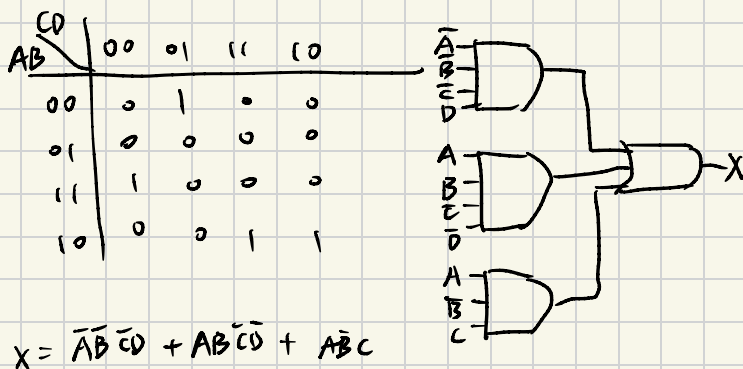
$X = ABC + A\bar{B}\bar{C} + A\bar{B}C$



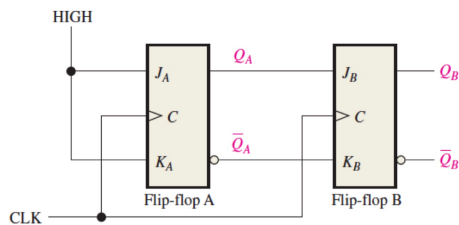
8. The input waveforms in Figure 6-72 are applied to a 2-bit adder. Determine the waveforms for the sum and the output carry in relation to the inputs by constructing a timing diagram.



19. You wish to detect only the presence of the codes 1010, 1100, 0001, and 1011. An active-HIGH output is required to indicate their presence. Develop the minimum decoding logic with a single output that will indicate when any one of these codes is on the inputs. For any other code, the output must be LOW.



24. For the circuit in Figure 7-89, determine the maximum frequency of the clock signal for reliable operation if the set-up time for each flip-flop is 3 ns and the propagation delays (t_{PLH} and t_{PHL}) from clock to output are 6 ns for each flip-flop.



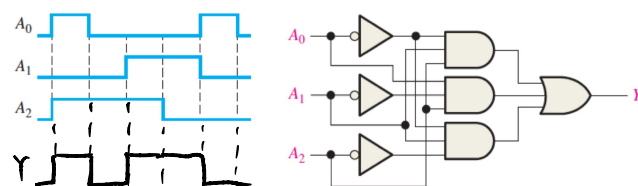
$$\begin{aligned}
 T &= t_{\text{set-up}} + t_{\text{comb}} + t_{\text{prop}} \\
 &= 3\text{ ns} + 0 + 6\text{ ns} = 9 \times 10^{-9}\text{ s} \\
 f_{\text{max}} &= \frac{1}{T} = 111.11 \times 10^6\text{ Hz}
 \end{aligned}$$

9. The following sequences of bits (right-most bit first) appear on the inputs to a 4-bit parallel adder. Determine the resulting sequence of bits on each sum output.

A_1	1010
A_2	1100
A_3	0101
A_4	1101
B_1	1001
B_2	1011
B_3	0000
B_4	0001

	t_1	t_2	t_3	t_4
A	1100	0001	1110	1011
B	1011	0010	0000	0011
Cin	0	1	0	0
S	1011	00100	01110	01110

20. If the input waveforms are applied to the decoding logic as indicated in Figure 6-76, sketch the output waveform in proper relation to the inputs.



$$Y = \bar{A}_0 A_1 A_2 + A_0 \bar{A}_1 A_2 + \bar{A}_0 A_1 \bar{A}_2$$